

Literature Status Note: CSP Subspaces of Spaces with Long Unconditional Bases

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Status

This is a **literature_already_answered** packet. It records an exact later-paper answer to an open problem extracted from Talponen’s 2008 paper; it is not claimed as a new proof produced in this run.

Source Question

The source is Jarno Talponen, *Lindelof type of generalization of separability in Banach spaces*, arXiv:0803.3541. In Section 3, Problem 3.3 (local source TeX line 375), the paper asks:

If X admits a long unconditional basis and $Y \subset X$ is a closed subspace with CSP, must Y be separable?

Supporting Answer

The supporting paper is Jarno Talponen, *Unconditional and bimonotone structures in high density Banach spaces*, arXiv:1604.04408. In the final remarks (local source TeX lines 895–896), the paper states that if a CSP space Y embeds into a Banach space X with a strong M-basis, then Y is separable, and explicitly adds that this settles Problem 3.3 of the 2008 paper.

Thus the answer to the extracted long-unconditional-basis problem is positive in the later literature. The supporting author explicitly identifies the target problem, so this belongs in **literature_already_answered** rather than as a new result or an agent-identified implication.

Scope Limits

This status note concerns only the long-unconditional-basis subspace question. It does not settle the other questions in arXiv:0803.3541 about finite sums, the general three-space problem for CSP, CSP together with the Radon–Nikodym property, or the existence of a CSP space without Corson’s property (C).

References

- J. Talponen, *Lindelof type of generalization of separability in Banach spaces*, arXiv:0803.3541.
- J. Talponen, *Unconditional and bimonotone structures in high density Banach spaces*, arXiv:1604.04408.